University of Florida, Department of Mathematics

Topology PhD Exam, January 2004

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that no covering of the 2-torus has the homotopy type of a figure eight.

2. Show that there is no retraction of the Möbius band onto its boundary.

3. Does there exist a covering space of the figure eight that has a non-trivial abelian fundamental group?

4. Show that the 2-torus with a point deleted does not admit a structure of a topological group.

5. Show that the space $R^2 \setminus A$ is path connected for any countable set A.

Answer the following with complete definitions or statements or short proofs.

6. State the Seifert–van Kampen Theorem.

7. State the Lefschetz Fixed Point Theorem.

8. State the Baire Category Theorem.

9. State the Arzelà–Ascoli Theorem.

10. Let X be a finite complex. Find $\chi(X \times S^n)$.

11. State the Five Lemma.

12. State the Poincaré Duality Theorem.

13. Define retraction and deformation retraction.

14. State the Mayer–Vietoris exact sequence for cohomology.

15. Let X be a complete metric space such that no point is isolated. Can X be countable?